

THE IMPACT OF CATEGORIZATION ON CONSUMPTION BEHAVIOR

Siyuan Yin

A DISSERTATION

in

Marketing

Presented to the Faculties of the University of Pennsylvania

in

Partial Fulfillment of the Requirements for the

Degree of Doctor of Philosophy

2025

Supervisor of Dissertation

Marissa Sharif, Associate Professor of Marketing

Graduate Group Chairperson

Eric Bradlow, K.P. Chao Professor of Marketing

Dissertation Committee

Cait Lamberton, Alberto I. Duran President's Distinguished Professor of Marketing

Maurice Schweitzer, Cecilia Yen Koo Professor of Operations, Information and Decisions

Barbara Mellers, George I. Heyman University Professor of Marketing and Psychology

Michael Norton, Harold M. Brierley Professor of Business Administration

To Ludwig Yin, for all the unconditional.

PREVIEW

ABSTRACT

THE IMPACT OF CATEGORIZATION ON CONSUMPTION BEHAVIOR

Siyuan Yin

Marissa Sharif

Consumers routinely make decisions about how to spend their resources, whether that are financial resources, such as money in checking accounts and gift cards, or the resource of time. For example, they may have to decide whether to spend their money on an item they might want but not need now or wait to spend their money on something else more essential later. Or they may have to decide whether to save their money for a savings goal or spend their money on something else they like now. In the context of time, people may have to decide whether to spend their time to continue working on a project or choose to take a break and browse social media. In my dissertation, I demonstrate the impact of categorization of resources, such as financial resources and time, on how consumers decide to spend and save these resources. To do so, I will discuss three essays. The first essay focuses on how a used (e.g., \$10 remaining on a gift card that originally had \$100 vs. unused, e.g., \$10 remaining on a gift card that originally had \$10) account affects how consumers spend financial resources. The second essay focuses on when, within a categorized time period, consumers prefer to spend time on leisure activities; for example, when it is close to the end of an hour (e.g., 1:50 PM) vs. when it is far away from the end of an hour (e.g., 2:00 PM). The third essay focusing on how different types of savings goals motivate consumers to save for the future, such as experiential goals (e.g., saving for hiking trips) vs. material goals (e.g., saving for hiking gear).

TABLE OF CONTENTS

ABSTRACT	iii
LIST OF TABLES	v
LIST OF FIGURES	vi
CHAPTER 1: HOW AND WHEN DOES A USED (VS. UNUSED) ACCOUNT AFFECT CONSUMPTION BEHAVIOR?	1
BIBLIOGRAPHY	38
CHAPTER 2: THE END-OF-PERIOD EFFECT: TEMPORAL CATEGORY BOUNDARIES INFLUENCE LEISURE ENGAGEMENT	43
BIBLIOGRAPHY	65
CHAPTER 3: THE LIFE YOU SAVE (FOR): EXPERIENCES DOMINATE GOODS IN MOTIVATING SAVINGS	71
BIBLIOGRAPHY	104
APPENDIX A SUPPLEMENTAL MATERIALS FOR HOW AND WHEN DOES A USED (VS. UNUSED) ACCOUNT AFFECT CONSUMPTION BEHAVIOR?	109
I: Chapter 1 Summary of Results	109
II: Chapter 1 Additional Analyses from Pre-registrations	110
III: Chapter 1 Supplemental Studies	113
APPENDIX B SUPPLEMENTAL MATERIALS FOR THE END-OF-PERIOD EFFECT: TEMPORAL CATEGORY BOUNDARIES INFLUENCE LEISURE ENGAGEMENT?	125
I. Chapter 2 Study 1 ATUS Activity Classification and Additional Analysis	125
II. Chapter 2 Study 3 Additional Measures.....	130
APPENDIX C SUPPLEMENTAL MATERIALS FOR THE LIFE YOU SAVE (FOR): EXPERIENCES DOMINATE GOODS IN MOTIVATING SAVINGS	131
I. Chapter 3 Supplemental Study A Saving Versus Borrowing	131
II. Chapter 3 Pretest of Saving Goals.....	134
III. Chapter 3 Study 6 Additional Analyses of Saving Behavior.....	136

LIST OF TABLES

Table 1. The Number of Participants Who Held onto Reward Points at Each Period Across Conditions	27
Table 2. The Impact of Relative Start Time on Leisure Activities Crowdsourced Categories.....	50
Table 3. Mean, Standard Deviation, And Regression Results of Mechanism Measures.....	88
Table 4. Goal Amount and Goal Length of Saving Goals by Goal Type	96

PREVIEW

LIST OF FIGURES

Figure 1. Study 1: The Likelihood of Spending \$5 to Purchase the Tasty Drink	8
Figure 2. Study 2A: A. The Likelihood of Spending \$20; B. The Valuation of the \$100 in a Checking Account	11
Figure 3. Study 2B: A. The Likelihood of Spending the 30,000 Points on the Running Shoes; B. The Valuation of the 30,000 Points.....	15
Figure 4. Study 3: The Likelihood of Spending the Money of the Gift Card Across Conditions	18
Figure 5. Study 4: A. The Likelihood of Spending the 30,000 Points from Credit Card Reward Programs. B. The Valuation of the 30,000 Points.....	24
Figure 6. Incentive-Compatible Studies: A. Study 5 (Holding onto Endowed Points). B. Study 6 (Donating Earned Points)	27
Figure 7. Study 6: Two Exemplar Trials of the Counting Task.....	29
Figure 8. Meta-Analysis Across Studies.....	34
Figure 9. Study 4: A. Likelihood of Spending Time on Social Media; B Valuation of the Time	57
Figure 10. Study 5B Stimuli	60
Figure 11. Studies 2A-2C: Saving Intentions Across Goal Types.....	80
Figure 12. Study 4: Mechanisms.....	89
Figure 13. Study 5: Saving Motivation and Perceived Versatility	92
Figure 14. Study 5: A Moderating Role of Goal Specificity	93
Figure 15. Study 6: Proximity To Success by Goal Type.....	96
Figure 16. Study 6: Average and Maximal Streaks by Goal Type	99

CHAPTER 1: HOW AND WHEN DOES A USED (VS. UNUSED) ACCOUNT AFFECT CONSUMPTION BEHAVIOR?

Yin, S., & Sharif, M. A. (2024). How and when does a used (vs. unused) account affect consumption behavior? *Journal of Experimental Psychology: General*, 153(4), 939–956. <https://doi.org/10.1037/xge0001541>

Siyuan Yin served as lead for conceptualization, data curation, formal analysis, investigation, methodology, visualization, writing—original draft, and writing—review and editing.

Imagine that you have a \$100 Best Buy gift card. Last month, you spent \$90 of this gift card and have \$10 remaining on it. As you are checking out, you find some new products that you might be interested in, such as a new iPhone case, costing \$8. While you have an iPhone case that is in good condition, this case has a unique design that you like. How likely would you be to purchase this iPhone case with your gift card? Now imagine that, instead of having \$10 remaining on a \$100 Best Buy gift card, you have an unused \$10 Best Buy gift card. Similarly, you find the new iPhone case, costing \$8. In this situation, would you be equally, more, or less likely to purchase the iPhone case than in the previous situation?

Individuals frequently have to make the choice to spend their resources now or, alternatively, to hold onto their resources for something else later. In some situations, the account they are considering spending from is used (e.g., some money of the gift card has been used) or unused (e.g., no money of the gift card has been used yet). In this research, we examine whether people are more (vs. less) likely to spend their resources when spending from a used account than otherwise equivalent resources from an unused account, holding constant the absolute amount of resources in the account.

We predict that people will be more likely to spend their remaining resources from a used account than otherwise equivalent resources from an unused account. People's judgments are regularly based on relative rather than absolute standards and are often influenced by the surrounding context (e.g., Baird et al., 1980; Campbell et al., 1958; Garner, 1954; Laming, 1997; Morewedge et al., 2007; Sharif & Oppenheimer, 2016; 2021; Sherman et al., 1978; Stewart & Brown, 2004; Stewart et al., 2002). Building on this research, we suggest that people are likely to engage in a within-account relative comparison, comparing how much they have left in their account to what they had *originally*. As a result, people value their remaining resources less in an account that is used than those in an

account that is unused. This decreased valuation of resources leads people to be more likely to spend them (Arkes et al., 1994; Bodkin, 1959).

We demonstrate this effect across seven studies ($N = 8,667$) in multiple domains, including endowed (e.g., gift cards and shopping reward points) and earned resources (e.g., checking accounts). Further, we examine the boundary conditions of this effect, revealing that the relative amount left in a used account moderates the effect of a used (vs. unused) account.

Theoretical Background

Division of Resources Influences Spending

Although many resources have a linear construct, for instance, time elapses and people spend money continuously, people often mentally or physically divide their resources, affecting their judgments of these resources and the resulting consumption behavior. Prior research has found that these divisions, whether mental or physical, largely affect how people spend their resources (Cheema & Soman, 2006; Heath & Soll, 1996; Sharif & Woolley, 2020; Sussman & Alter, 2012; Sussman & O'Brien, 2016; Thaler, 1985; 1999; Thaler & Sunstein, 2008).

In terms of mental divisions of resources, mental accounting research has found that individuals frequently label or “ earmark ” their resources for different needs, designating these accounts as for “ education ” only or for “ food ” only. Earmarking resources can act as a budgeting mechanism, such that people attempt to only spend their resources in the manner in which they have been “ earmarked ” or “ labeled, ” increasing self-control (Heath & Soll, 1996; Sharif & Shu, 2017; Shefrin & Thaler, 1988; Thaler, 1985). Further, categorization cues, such as arbitrary labels, can externally lead consumers to divide their resources such as time (Rajagopal and Rha 2009; Tu and Soman 2014), products (LeClerc, Hsee, and Nunes 2005), and space (Mishra and Mishra 2010; Zhao, Lee, and Soman 2012).

Physical divisions of resources can also affect people’s consumption behavior. Partitioning an aggregate quantity into smaller units reduces the amount that people consume (Cheema & Soman, 2008). For example, people ate fewer chocolates when each chocolate was individually wrapped (vs. not wrapped) in a box. The authors suggest that the physical nature of the partition drives this slowed consumption behavior. That is, before consuming each chocolate, individuals must physically open the wrapper. This small transaction cost draws attention to the partition and adds a temporal delay before consuming it, leading individuals to consume less when resources are partitioned into smaller units (Cheema & Soman, 2008; Soman & Cheema, 2011; Soman et al., 2010).

These divisions often create a mental or physical grouping of a set of resources, what we define as an “account” in this paper. For example, a gift card can be an account for money, and a credit card reward account can be an account for reward points. In particular, we examine how spending from a used account rather than from an unused account can affect future consumption behavior, holding the absolute amount of resources constant. We suggest that when an account is used and thus has relatively less remaining, people are more likely to spend their remaining resources than when it is unused and has relatively more remaining. In doing so, we are the first to empirically examine how the relative amount remaining in an account affects future consumption behavior, when, and why.

Different Valuation and Spending of Resources in A Used vs. Unused Account

Prior research has found that individuals often make relative judgments about stimuli rather than absolute judgments (Hsee, 1996; 1998; Hsee & Leclerc, 1998; Hsee et al., 2013; Morewedge et al., 2007; Sharif & Oppenheimer, 2016; 2021; Sherman et al., 1978; Stewart et al., 2002). For example, how expensive a car is perceived to be depends more on the set of cars that are being evaluated at the same time than the objective price of the car itself. If the other cars being evaluated at the same time are less expensive than the target car, then the target car will be perceived as expensive. However, if the other cars are more expensive, the target car might be perceived as a good deal.

As demonstrated in this example, consumers’ judgments are largely affected by which reference points they rely on. Indeed, consumers’ judgment of a discount depends on whether they focus on the topical (vs. absolute) frame (Darke & Freedman, 1993; Tversky & Kahneman, 1981). For example, saving \$5 on a \$15 calculator can be considered a better deal than saving \$5 on a \$125 jacket when focusing on the relative amount saved off the product (e.g., saving 33% off the calculator vs. 4% off the jacket) (see also Heath et al., 1995). However, when focusing on the absolute frame, consumers may perceive the discount similarly, saving an equivalent of \$5 in both cases (see also Darke & Freedman, 1993).

Further, people often rely naturally on different reference points depending on the choice environment. For example, when evaluating one product in isolation, people tend to make a judgment about the product based on its attributes that are easy to evaluate, whereas when comparing two products in the same category, people make a judgment about the product based on the other product as a reference (Hsee, 1996; 1998; Hsee & Leclerc, 1998;

Hsee et al., 2013). As a result, individuals often have different preferences depending on whether they evaluate two different products separately or jointly (Hsee, 1996; 1998; Hsee & Leclerc, 1998; Hsee et al., 2013).

Building on this research, we suggest that when assessing their resources in a used account rather than in an unused account, people naturally rely on the original amount *in* the account as a reference point. That is, they engage in a within-account comparison. Although they evaluate their account in isolation (i.e., they are not comparing one account to another separate account), they make a relative judgment *within* the account itself, comparing how much of their account they have remaining relative to how much of the account *they had originally*.

We suggest this comparison leads consumers to value the remaining resources less. That is, by comparing the larger original amount of resources to the smaller remaining amount, consumers feel that their remaining resources are less valuable in a used account (e.g., memory-based norm, Bordalo et al., 2020; accessible mental accounts, Morewedge et al., 2007; contexts of social issues, Sherman et al., 1978; categorization of tone and geometric figures, Stewart et al., 2002). Specifically, we operationalize value based on the definition of “psychological value,” which is the perception of the importance, worth, or usefulness of an item (Cohen et al., 2022). Thus, we suggest that people may perceive their remaining resources to feel less important, hold less worth, and feel less useful in a used account than in an unused account.

There are many reasons why a within-account comparison might lead people to devalue their resources in a used (vs. unused) account. For one, comparing the larger original amount of resources to the smaller remaining amount of resources in a used account might make those remaining resources feel smaller (Morewedge et al., 2007; Stewart et al., 2002) and thus less useful for future expenses and consumption (Gourville, 1998; Mishra et al., 2006; Raghubir & Srivastava, 2009). Second, people may make different inferences about themselves and their consumption behavior if an account is used (vs. unused) (e.g., Kardes et al., 2004), such that they infer they value the resources less if they are in a used account rather than in an unused account. Third, people may perceive that they have achieved some type of earmarked goal when a large proportion of their account has been used (e.g., spent \$90 on a pair of New Balance shoes with a \$100 New Balance gift card), and thus perceive the remaining resources to be extra, and thus less valuable (e.g., Dhar & Simonson, 1999; Fishbach & Dhar, 2005; Kivetz & Simonson, 2002).

Because people value their resources less in a used account, we suggest that they are more likely to spend them. Indeed, a substantial body of research suggests that consumers’ likelihood of spending a resource depends on

how much they value the resource (Arkes et al., 1994; Bodkin, 1959; Carlsson et al., 2013; Frederick et al., 2009; Heilman et al., 2002; Muehlbacher & Kirchler, 2009; Spiller 2011; Stilley et al., 2010). For example, people spend windfall gains (a gain that may be less valued) more readily than earned gains (e.g., more likely to spend a tax rebate vs. savings to make a risky investment, and more likely to spend a gain obtained from a lottery vs. anticipated earnings on a portable TV instead of saving the money in the bank; Arkes et al., 1994; Bodkin, 1959). Relatedly, shoppers increase spending on both planned and unplanned items when they have slack in their mental budget (e.g., savings from coupons for planned items, Heilman et al., 2002; Stilley et al., 2010). On the other hand, considering high-value opportunity costs, or high-value alternative uses for one's resource, leads people to be less likely to spend resources (Frederick et al., 2009; Spiller, 2011).

We will begin by examining the main effect, comparing an unused account, an account that has 100% remaining, to a used account, an account that has a small percentage left (e.g., 20% remaining) across several domains, including gift cards, checking account, credit card reward points, and incentive-compatible decisions. Importantly, we hold constant the absolute total amount of remaining resources across accounts. After establishing the main effect, we will then reveal at a continuous level that as the relative amount remaining in the used account *decreases*, people are increasingly *more* likely to spend the resources in the account. In doing so, we will demonstrate that the effect of a used (vs. unused) account on spending is attenuated if there is a relatively large amount remaining in the used account.

Overview of Studies

Across seven experimental studies ($N = 8,667$; six preregistered), we examine the effect of a used (vs. unused) account on the likelihood of spending resources. In Study 1 and Supplementary Study A, we provide evidence for the main effect in the context of gift cards and checking accounts and reveal that the effect is due to a within-account comparison, rather than due to a comparison to any large reference point. Studies 2A and 2B reveal that people value their remaining resources less in a used account than those in an unused account, leading them to be more likely to spend them. Importantly, in Studies 3 and 4, we demonstrate that the proportion of the account remaining moderates the effect. That is, if a used account is relatively full, the effect is attenuated. To test the robustness of the effect, Study 5 replicates the effect in an incentive-compatible behavioral experiment of online shopping. Finally, Study 6 and Supplementary Study C generalize the effect with both earned and endowed

resources from spending to donation decisions. We summarize the main results of the studies in Appendix A (Table S1).

Study Samples and Constraints on Generality

In all of our studies, we determined the sample size in advance based on the effect size from pilots of the studies, and we reported all of our measures, manipulations, exclusions, and rules for determining sample size. All studies except for Study 5 were preregistered on AsPredicted.org, and the links to those preregistrations are in the method section of each study. We followed all of our preregistered analysis plans and exclusion rules and noted any exceptions. The research was approved by the University of Pennsylvania's Institutional Review Board (Protocol #828347).

In each experimental study, we recruited participants on Amazon Mechanical Turk or Prolific Academic in the U.S., which has a more diverse population than a typical American college with respect to age, gender, race, education, and employment (Peer et al., 2017). We tested the effect of a used (vs. unused) account on spending across multiple consumption domains, with both endowed and earned resources, such as gift cards, credit card reward programs, checking accounts, and shopping rewards. We also generalized the effect of a used (vs. unused) account from spending to charitable giving. The generality of our results is limited to internet users with some connection to the United States—we primarily studied the effect with American participants, and so our results may not generalize in different countries with different languages and cultures.

Transparency and Openness

We take steps to ensure the transparency and openness of our study design and analyses. In our OSF repository, we include (1) the complete materials that show the full details of each study, (2) data for our experiments, and (3) analysis syntax that outlines the processing of the data for all studies. All are available at OSF: https://osf.io/uh2dr/?view_only=f1b31ef057144c08ad598fce9f651a2f.

Study 1: Heightened Proclivity to Spend a Used (vs. Unused) Gift Card

Study 1 aims to provide initial evidence for the effect of a used (vs. unused) account on spending decisions with gift cards. We suggest that people will be more likely to spend their remaining resources from a used account than otherwise equivalent resources from an unused account. In particular, we propose that the effect is due to individuals comparing the amount that they have in their account now to the amount that they had originally. This

within-account comparison results in individuals feeling that their remaining resources are less valuable in a used account than in an unused account, leading them to be more likely to spend them.

However, an alternative explanation is that people simply compare their remaining resources to a larger reference point in a used account (which is not present in an unused account), leading them to similarly devalue their remaining resources (Morewedge et al., 2007; Sherman et al., 1978; Stewart et al., 2002). That is, a *within*-account relative comparison may not be essential to our theory. If this alternative explanation is true, the addition of a larger reference point outside an unused account should lead people to be as likely to spend their resources as people with a used account. However, if, instead, we still find that people are more likely to spend from a used account than from an unused account, regardless of the presence of a larger reference point outside the unused account, we can rule out this alternative explanation. We will test this alternative account in this study.

Method

This study was pre-registered (https://aspredicted.org/52X_HZ4) for 1200 participants on Prolific Academic. One thousand two hundred three participants completed this study and were randomly assigned to one of the three between-subject conditions: a used account, an unused account with a greater reference point, vs. an unused account without a greater reference point. In the *used account* condition, participants imagined that they had received a \$40 gift card to a clothing store from one of their friends and spent \$32 of this gift card last month. In the *unused account without a greater reference point* condition, participants imagined that they had received an \$8 gift card to a clothing store from one of their friends. In the *unused account with a greater reference point* condition, participants imagined that they had received a \$32 gift card to a clothing store from one of their friends. Last month they spent \$32 of this gift card and no longer had any money on the gift card. They also received an \$8 gift card to the same clothing store from one of their friends.

All participants then imagined that they visited the clothing store, and as they were checking out, they saw that the clothing retailer was selling some tasty drinks (e.g., smoothie, latte). All the drinks cost \$5, and they could use their gift card to pay for it. Participants were then asked how likely they would be to buy a tasty drink with their gift card on a 0-100 slider from 0 (*Not at all likely*) to 100 (*Very likely*). At the end of the study, we asked participants their age (“What is your age?” [Text-entry]) and gender (“What is your gender?” Male, Female, Other, and Prefer not to answer).

Results¹

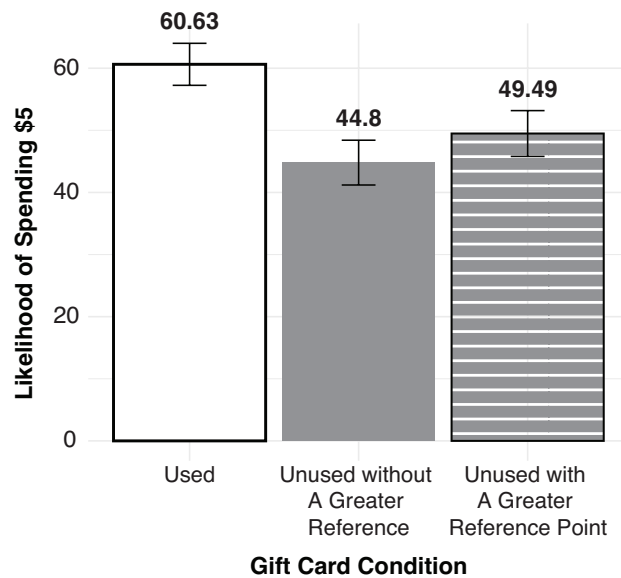
As preregistered, we excluded ten participants who failed the attention check, so data were analyzed with the remaining 1193 individuals ($M_{\text{age}} = 39.50$, $SD = 12.86$, $\text{Range} = [19, 93]$; $N_{\text{Female}} = 594$, $N_{\text{Male}} = 584$, $N_{\text{other}} = 9$, $N_{\text{Undisclosed}} = 6$).

We conducted a linear regression predicting the likelihood of spending \$5 on a tasty drink from two dummy variables representing the account conditions with the used account condition as a reference group. The results were as predicted: participants were more likely to spend \$5 on the tasty drink in the used account condition than those in the unused account without a greater reference point condition ($M_{\text{used}} = 60.63$, $SD = 34.45$ vs. $M_{\text{unused without a greater reference point}} = 44.80$, $SD = 36.91$, $d = .44$; $b_{\text{unused without a greater reference point}} = -15.82$, $SE = 2.54$, $t(1190) = -6.22$, $p < .001$, 95% CI of the difference = $[-20.81, -10.84]$, $\beta = -.20$). Importantly, participants were also more likely to spend \$5 on the tasty drink in the used condition than those in the unused with a greater reference point condition ($M_{\text{used}} = 60.63$, $SD = 34.45$ vs. $M_{\text{unused with a greater reference point}} = 49.49$, $SD = 36.88$, $d = .31$; $b_{\text{unused with a greater reference point}} = -11.14$, $SE = 2.57$, $t(1141) = -4.33$, $p < .001$, 95% CI of the difference = $[-16.19, -6.09]$, $\beta = .14$). Further, the likelihood of spending differed only marginally between these two unused account conditions ($d = .13$, $p = .068$, 95% CI of the difference = $[-.35, 9.72]$).

Figure 1. Study 1: The Likelihood of Spending \$5 to Purchase the Tasty Drink²

¹ We included additional preregistered analyses in the Supplementary Materials for all studies.

² All error bars in figures are 95% confidence intervals.



Discussion

Study 1 provides initial evidence for the effect of a used (vs. unused) account that people are more likely to spend their resources in a used account than otherwise equivalent resources in an unused account in the context of gift cards. Further, we reveal that this is mainly driven by a within-account comparison, comparing the remaining resources with what the account originally had, instead of any greater reference point. To show the generalizability of the effect in a different context—checking accounts where the resources are often earned instead of endowed, we conducted Supplemental Study A to examine whether people would be more likely to spend their remaining resources from a used checking account rather than an unused checking account. We replicated the same pattern that people were more likely to spend \$20 to order some takeout from a food truck with a used checking account than with an unused checking account. The findings of Study 1 and Supplemental Study A suggest that the effect holds for both endowed and earned resources.

Study 2A: Spending of Resources from a Used, Unused, versus Unspecified Account

Many accounts are often used frequently, such as checking accounts. In this study, we aimed to examine how people spend their resources from a frequently used account, such as a checking account, when the original amount (e.g., the total amount at the beginning of each month) of the account *is not* specified. That is, people are simply told how much is in the account currently, as is often the status quo with checking accounts. We expected that people would be less likely to spend from a checking account when the original amount specified indicated the

account was *unused* than when the original account is not specified, as people would naturally encode this account as used.

Further, in this study, we further examine the mechanism behind this effect. In particular, we suggest that due to a within-account comparison, people value their resources less in a used account than in an unused account. We assess whether people do indeed value their resources less in a used account, as we predict, and further whether this mediates the effect on spending behavior.

Method

This study was preregistered (https://aspredicted.org/YW2_FXY) for 1,200 Prolific participants. One thousand two hundred and two participants completed the study and were randomly assigned to one of three between-subject conditions: a used, unused, vs. unspecified checking account. All participants learned that they kept the majority of their money in their savings account and primarily used their checking account for smaller purchases. Specifically, in the *used checking account* condition, participants had \$500 in their checking account, spent \$400 earlier this month on a few things that they liked, and had \$100 available in their checking account. In the *unused checking account* condition, participants had two checking accounts at the same bank. They had \$400 in one checking account, Checking Account A, spent \$400 earlier this month on a few things they liked, and had \$0 remaining in Checking Account A. They also had \$100 in another checking account, Checking Account B, at the same bank, spent \$0 from Checking Account B this month so far, and still had \$100 available in Checking Account B. In the *unspecified checking account* condition, participants were simply told that they had \$100 in their checking account. Earlier this month they spent \$400 with their credit card on a few things that they liked.

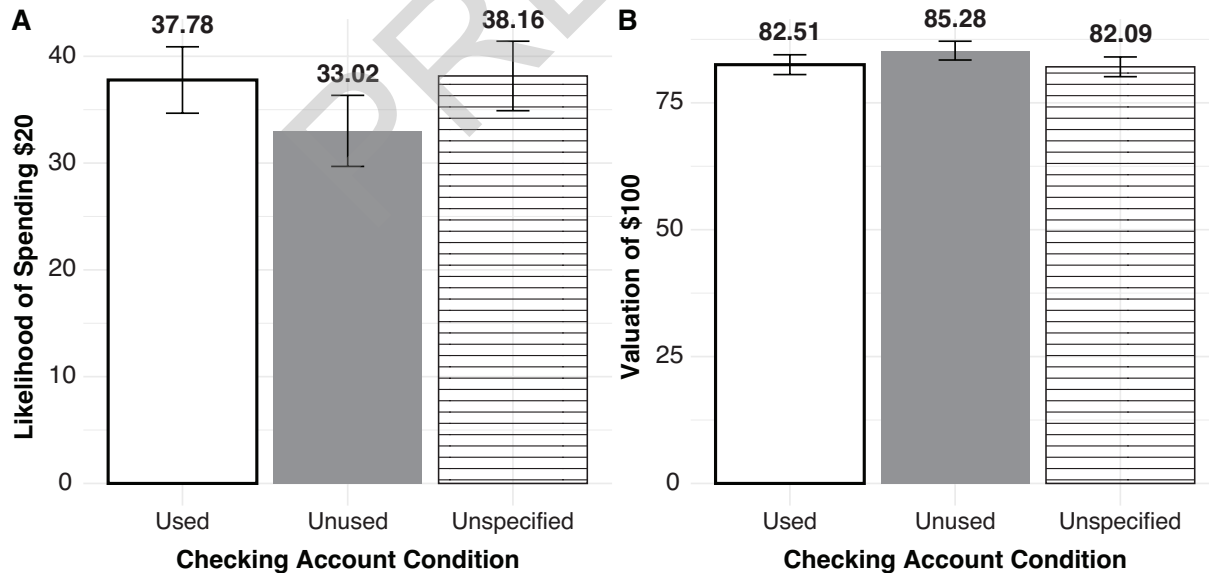
All participants then read that they came across one of their favorite food trucks nearby that only accepts cash. The meals cost on average \$20 including tips. They could take out \$20 from their checking account to pay for the meal. They had some leftovers at home that they could warm up, but they also really liked the food from the food truck. Participants were then asked (1) how likely they would be to spend \$20 from their checking account to order a meal from the food truck on a 0-100 slider from 0 (*Not at all likely*) to 100 (*Very likely*), and (2) prior to making their decision to make an order, how much they would value the money in their checking account on a 0-100 slider from 0 (*Not at all value*) to 100 (*Value a lot*). At the end of the study, we asked participants their age (“What is your age?” [Text-entry]) and gender (“What is your gender?” Male, Female, and Prefer not to answer).

Results

As preregistered, we excluded twelve participants who failed the attention check, so data were analyzed with the remaining 1190 individuals ($M_{\text{age}} = 39.96$, $SD = 13.82$, $\text{Range} = [18, 82]$; $N_{\text{Female}} = 582$, $N_{\text{Male}} = 582$, $N_{\text{other}} = 16$, $N_{\text{Undisclosed}} = 10$).

We first conducted a linear regression to regress the spending likelihood on two dummy variables representing the checking account conditions. Replicating the findings in previous studies, we found that participants were more likely to spend \$20 from a used checking account than from an unused checking account ($M_{\text{used}} = 37.78$, $SD = 31.64$ vs. $M_{\text{unused}} = 33.02$, $SD = 33.77$, $d = .15$; $b_{\text{unused}} = -4.76$, $SE = 2.32$, $t(1187) = -2.06$, $p = .040$, 95% $CI = [-9.31, -.22]$, $\beta = -.069$). We then compared the spending likelihood between the unspecified checking account and the other two account conditions. Participants were less likely to spend from an unused checking account than from an unspecified checking account ($M_{\text{unused}} = 33.02$, $SD = 33.77$ vs. $M_{\text{unspecified}} = 38.16$, $SD = 32.77$, $d = .15$; $b_{\text{unused}} = -5.14$, $SE = 2.33$, $t(1187) = -2.21$, $p = .027$, 95% $CI = [-9.71, -.57]$, $\beta = -.074$). Further, participants were equally likely to spend from a used checking account and from an unspecified checking account ($M_{\text{used}} = 37.78$, $SD = 31.64$ vs. $M_{\text{unspecified}} = 38.16$, $SD = 32.77$, $d = .01$; $b_{\text{unspecified}} = .38$, $SE = 2.33$, $t(1187) = .16$, $p = .87$, 95% $CI = [-4.19, 4.94]$, $\beta = .005$).

Figure 2. Study 2A: A. The Likelihood of Spending \$20; B. The Valuation of the \$100 in a Checking Account



We next examined the valuation of the money in these accounts with a linear regression from two dummy variables representing the checking account conditions. Consistent with our theorizing, participants perceived the money in a used checking account as less valuable than that in an unused checking account ($M_{\text{used}} = 82.51$, $SD = 19.86$ vs. $M_{\text{unused}} = 85.28$, $SD = 18.81$, $d = .14$; $b_{\text{unused}} = 2.77$, $SE = 1.38$, $t(1187) = 2.02$, $p = .044$, 95% CI = [.074, 5.47], $\beta = .067$). We also compared the valuation between the unspecified account and the other two account conditions. Participants valued their money in an unused checking account more than that in an unspecified checking account ($M_{\text{unused}} = 85.28$, $SD = 18.81$ vs. $M_{\text{unspecified}} = 82.09$, $SD = 19.58$, $d = .17$; $b_{\text{unused}} = 3.19$, $SE = 1.38$, $t(1187) = 2.31$, $p = .021$, 95% CI = [.48, 5.90], $\beta = .077$). Further, participants valued their money similarly in a used checking account and in an unspecified checking account ($M_{\text{used}} = 82.51$, $SD = 19.86$ vs. $M_{\text{unspecified}} = 82.09$, $SD = 19.58$, $d = .02$; $b_{\text{unspecified}} = -.42$, $SE = 1.38$, $t(1187) = -.31$, $p = .76$, 95% CI = [-29.13, 2.9], $\beta = -.010$).

We conducted a mediation analysis using the bootstrap method with 10,000 samples to test our proposed process that the devaluation of available resources leads to a greater likelihood of spending in the used account condition than in the unused account condition (Model 4; Hayes, 2013). Specifically, we predicted that the devaluation of the money in a used checking account would increase the spending likelihood, in comparison to that in an unused checking account. Consistent with our hypothesis, valuation significantly mediated the effect of the unused (vs. used) checking account condition on spending: $a \times b = -.91$, $SE = .49$, 95% CI = [-1.96, -.025]. Furthermore, valuation also significantly mediated the effect in the unused (vs. unspecified) checking account condition on spending: $a \times b = -1.47$, $SE = .66$, 95% CI = [-2.77, -.18]).

Discussion

We first replicate the effect of a used (vs. unused) checking account on spending decisions as in previous studies. Second, we find that people's spending likelihood with an unspecified checking account is similar to a specified used checking account and greater than a clearly unused checking account. The findings provide evidence that people naturally perceive a checking account as a used account, and thus are more likely to spend money from that account than from an unused account. This provides implications for the real world in terms of how to decrease spending with checking accounts (i.e., framing accounts as unused), which we discuss in greater detail in the General Discussion. While we find with checking accounts that people naturally consider an unspecified account to

be a used account, other accounts that are unspecified might be thought of as unused, if they are not used frequently, or consumers forget that they have used them before.

Study 2B: The Role of Within-Account Comparisons

We have illustrated evidence supporting the effect of a used (vs. unused) account in the context of gift cards and checking accounts. In the next study, we aim to further demonstrate evidence for our effect in the context of credit card reward programs, as well as additional evidence for our mechanism. To do so, we will use a three-cell between-subject design, as in Study 1, to demonstrate that indeed people devalue the resources in a used account due to a *within*-account comparison.

Further, we will examine whether the effect holds in a common context where people often need to choose to spend between cash and reward points in their credit card program. If people have points available to spend, they should be more likely to spend these points than cash, as reward points are considered more of a windfall gain (Arkes et al., 1994; Bodkin, 1959; Carlsson et al., 2013). However, if these reward points are from an unused account, consumers may be less likely to spend these points, and use their cash instead. Thus, we expected to find the effect when people choose to spend resources from a used (vs. unused) account versus cash. We test this when holding the alternative (i.e., cash) constant across account conditions.

Method

This study was pre-registered (<https://aspredicted.org/blind.php?x=h8ch58>) for 1200 participants on Amazon Mechanical Turk (MTurk). One thousand one hundred and ninety-seven MTurk participants completed this study and were randomly assigned to one of the three between-subject conditions: a used account, an unused account with a greater reference point, vs. an unused account without a greater reference point. In the *used account* condition, participants imagined that they had accumulated 100,000 points in a credit card reward program, had spent 70,000 points earlier this year, and now had 30,000 points available. In the *unused account without a greater reference* condition, participants imagined that they had accumulated 30,000 points in a credit card reward program, had spent 0 points this year so far, and now still had 30,000 points available. In the *unused account with a greater reference point* condition, participants imagined that they had two credit cards – Credit Card A and Credit Card B. They had accumulated 70,000 points in Credit Card A, spent 70,000 of these points, and no longer had any points available in Credit Card A. They had also accumulated 30,000 points in Credit Card B, had spent 0 of these points

so far, and still had 30,000 points available in Credit Card B. Importantly, they were told that the points in both programs could be applied to the *same rewards*.

All participants were then asked: “How much would you value the 30,000 points?” on a 0-100 slider from 0 (*Not at all valuable*) to 100 (*Very valuable*)³. Following this, they imagined that they received an email advertisement that they could use these 30,000 points to buy some running shoes. Their current running shoes were still in good condition, but the new running shoes were lighter than their current running shoes. Next, participants were asked, “Would you use your points to buy these running shoes or instead use your cash and save your points for a different reward later?” on a 0-100 slider from 0 (*Very likely to use cash to buy these running shoes*) to 100 (*Very likely to use points to buy these running shoes*). At the end of the study, we asked participants their age (“What is your age?” [Text-entry]) and gender (“What is your gender?” Male, Female, and Prefer not to answer).

Results

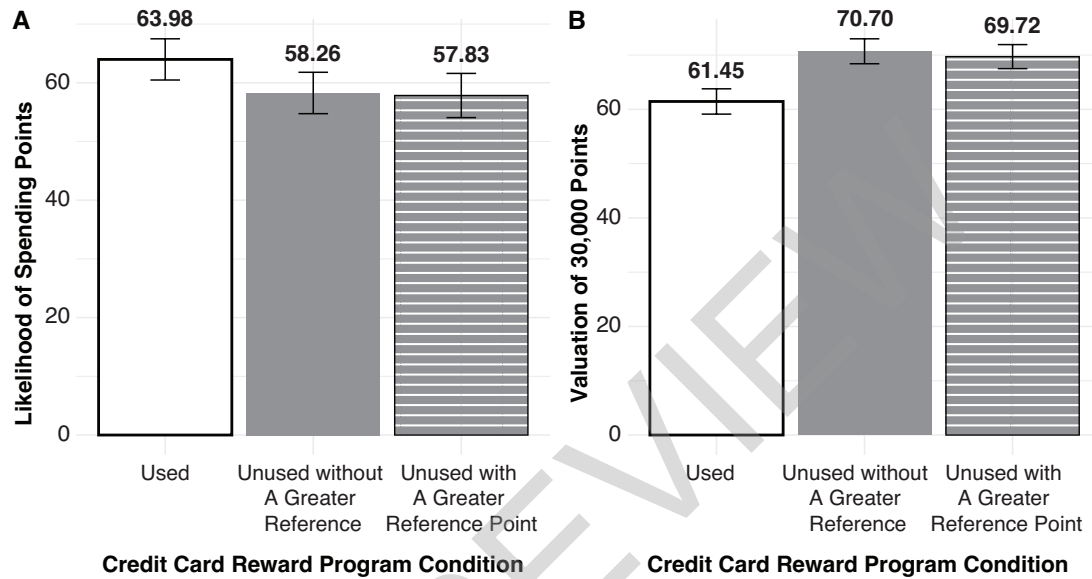
As preregistered, we excluded fifty-three participants who failed the attention check, so data were analyzed with the remaining 1144 individuals ($M_{\text{age}} = 32.03$, $SD = 11.32$, $\text{Range} = [18, 78]$; $N_{\text{Female}} = 611$, $N_{\text{Male}} = 521$, $N_{\text{Undisclosed}} = 12$).

We first conducted a linear regression predicting the likelihood of spending the 30,000 reward points from two dummy variables representing the account conditions with the used account condition as the reference group. The results were as predicted: participants were more likely to spend the 30,000 reward points on the running shoes in the used account condition than those in the unused account without a greater reference point condition ($M_{\text{used}} = 63.98$, $SD = 34.98$ vs. $M_{\text{unused without a greater reference point}} = 58.26$, $SD = 34.99$, $d = .16$; $b_{\text{unused without a greater reference point}} = -5.72$, $SE = 2.59$, $t(1141) = -2.21$, $p = .027$, 95% CI of the difference = $[-10.79, -.64]$, $\beta = -.08$). Importantly, participants were also more likely to spend the 30,000 reward points on the running shoes in the used condition than those in the unused with a greater reference point condition ($M_{\text{used}} = 63.98$, $SD = 34.98$ vs. $M_{\text{unused with a greater reference point}}$

³ We used a similar design to understand how consumers perceive valuation (Supplemental Study B). 387 participants were randomly assigned to one of two conditions: used vs. unused account. Participants were asked (1) “How much would you value the 30,000 points?” (0 = *Not at all valuable*; 100 = *Very valuable*), (2) “How large would the 30,000 points in your account feel?” (0 = *Very small*; 100 = *Very large*), (3) “How useful would the 30,000 points in your account feel?” (0 = *Not at all*; 100 = *Very useful*), (4) “How important would the 30,000 points in your account feel?” (0 = *Not at all*; 100 = *Very important*), and (5) “How attached would you feel to the 30,000 points in your account?” (0 = *Not at all*; 100 = *Very attached*). The valuation measure was correlated with all other four measures: all r s $\geq .62$, all p s $< .001$.

= 57.83, $SD = 37.27$, $d = .17$; $b_{\text{unused with a greater reference point}} = -6.15$, $SE = 2.59$, $t(1141) = -2.37$, $p = .018$, 95% CI of the difference = [-11.23, -1.07], $\beta = .08$). Further, the likelihood of spending the 30,000 reward points on the running shoes did not differ between these two unused account conditions ($d = .01$, $p = .88$, 95% CI of the difference = [-5.52, 4.66]).

Figure 3. Study 2B: A. The Likelihood of Spending the 30,000 Points on the Running Shoes; B. The Valuation of the 30,000 Points



We further examined if consumers' valuation of the reward points differed by condition. We conducted a linear regression predicting the valuation of 30,000 reward points from two dummy variables representing the account conditions with the used account condition as a reference group. As predicted, we found that participants valued the 30,000 reward points less in the used account condition than the unused account without a greater reference point condition ($M_{\text{used}} = 61.45$, $SD = 23.23$ vs. $M_{\text{unused without a greater reference point}} = 70.70$, $SD = 22.80$, $d = .40$; $b_{\text{unused without a greater reference point}} = 9.25$, $SE = 1.64$, $t(1141) = 5.64$, $p < .001$, 95% CI of the difference = [6.03, 12.46], $\beta = .19$). Importantly, participants also valued the 30,000 reward points less in the used condition than the unused with a greater reference point condition ($M_{\text{used}} = 61.45$, $SD = 23.23$ vs. $M_{\text{unused with a greater reference point}} = 69.72$, $SD = 21.92$, $d = .37$; $b_{\text{unused with a greater reference point}} = 8.26$, $SE = 1.64$, $t(1141) = 5.04$, $p < .001$, 95% CI of the difference = [5.04, 11.48],

$\beta = .17$). Consistent with our theory, the valuation of 30,000 reward points did not significantly differ between these two unused account conditions ($d = .04$, $p = .55$, 95% CI of the difference = [-4.21, 2.24]).

We conducted a mediation analysis using the bootstrap method with 10,000 samples to test our proposed process that devaluation of available resources leads to a greater likelihood of spending in the used account condition than in the unused account conditions (Model 4; Hayes, 2013). Specifically, we predicted that devaluation of the points would increase the likelihood of spending the 30,000 reward points on running shoes in the used account condition, in comparison to the unused without and with greater reference point conditions. Consistent with our hypothesis, valuation significantly mediated the effect in the used condition (vs. unused without a greater reference point condition, $a \times b = -1.30$, $SE = .51$, 95% CI = [-2.39, -.41]; vs. unused account with a greater reference point condition, $a \times b = -1.16$, $SE = .46$, 95% CI = [-2.15, -.36]).

Discussion

We replicated our results from previous studies that a used account leads to a greater likelihood of spending resources than an unused account. Further, while holding constant the absolute amount of resources across conditions, the resources in a used account were valued less than the resources in the unused account, with or without the presence of a greater reference outside the account. We also found consistent evidence for our proposed mechanism that the devaluation of resources mediates the effect of a used account on subsequent purchase behavior.

This study has important implications for stockpiling behavior. In particular, one factor that might increase consumers' likelihood to hold onto reward points is the perception of the account being unused. Often consumers must accrue reward points for a while before earning enough to spend them on a reward, leading the account to be unused for quite some time. They may finally reach the point at which they have enough rewards to redeem them for a reward. However, when they make a choice between spending from this unused reward account for the first time and spending their cash for a reward (e.g., an airline ticket), consumers might opt to use cash due to their resources appearing more valuable in this unused account.

Study 3: How Does the Proportion of the Account Remaining Moderate the Effect of a Used Account?

In Studies 1-2, we demonstrated that people are more likely to spend resources in a used account than in an unused account. In the next study, we examine if the relative amount remaining in the account moderates our effect. We suggest that people compare their lower remaining resources to the greater original amount in the used account

rather than in the unused account, perceiving the lower remaining resources to be less valuable in comparison to the greater original value of the account. If this theory holds, we should find that the lower the relative proportion remaining in the used account, the more people will devalue their resources and thus become more likely to spend them. On the other hand, if the relative proportion remaining in the used account is high, and thus there is not a big difference between the original amount and the remaining amount, people should be less likely to devalue their resources, and thus the effect should be attenuated. We tested this in Study 3.

In particular, in this study, we continuously manipulate the relative amount left in a used account, examining the impact of having 60%, 40%, and 20% left relative to an account that has 100% remaining, holding constant the absolute amount of resources in the account. We expected that people would be more likely to spend their resources the less they have relatively in their account. We measure and compare the likelihood of purchasing with a used account and with an unused account.

Method

This study was pre-registered (<http://aspredicted.org/blind.php?x=hp49ht>) for 1200 participants on MTurk. Twelve hundred and four MTurk participants completed this study and were randomly assigned to one of the 3 (account remaining: \$24, \$16, vs. \$8) x 2 (gift card condition: used vs. unused) between-subject conditions. In the *used gift card* conditions, participants were told that they had received one \$40 gift card to a clothing store from one of their friends. They had spent \$16, \$24, or \$32 of this gift card last month, and now had \$24, \$16, or \$8 remaining on the gift card, respectively. That is, the account had been used, and the proportion of the account remaining was 60%, 40%, and 20% in the used gift cards with \$24, \$16, and \$8 conditions, respectively. In the *unused gift card* conditions, participants were told that they received a \$24, \$16, or \$8 gift card to a clothing store from one of their friends in the \$24, \$16, and \$8 conditions, respectively. As these gift cards were unused, they all had 100% remaining in the account. To hold constant prior spending between the used and unused conditions, participants in the unused conditions were also told that prior to receiving this gift card, they had spent \$16, \$24, or \$32 at the same clothing store last month.⁴

⁴ To confirm that participants did not make different inferences about how expensive the items at the clothing store are, we ran a post-test ($N = 198$) examining the perceived expensiveness between the used and unused \$8 gift card conditions; we did not find significant differences between the used and unused \$8 gift card conditions ($M_{\text{used}} = 3.09$, $SD = 1.27$ vs. $M_{\text{unused}} = 2.99$, $SD = 1.09$; $t(192) = .60$, $p = .55$).

All participants were then asked, “As you are checking out, you see that the clothing retailer is selling some tasty drinks (e.g., smoothie, latte). All of the drinks cost \$5, and you can use your gift card to pay for a tasty drink. How likely would you be to buy the tasty drink now with your gift card?” on a 0-100 slider from 0 (*Not at all likely*) to 100 (*Very likely*).

At the end of the study, we asked participants their age (“What is your age?” [Text-entry]) and gender (“What is your gender?” Male, Female, and Prefer not to answer).

Results

As preregistered, we excluded seventy-five participants who failed the attention check, so data were analyzed with the remaining 1129 individuals ($M_{\text{age}} = 33.87$, $SD = 12.97$, $\text{Age} = [18, 78]$; $N_{\text{Female}} = 588$, $N_{\text{Male}} = 527$, $N_{\text{Undisclosed}} = 14$).

First, we assessed whether participants are more likely to spend their resources when the account has relatively less in it. We conducted a linear regression predicting the likelihood of spending from a dummy variable representing the used ($= 1$) vs. unused ($= 0$) account condition, a linear coding representing the account remaining condition ($\$24 = 1$, $\$16 = 0$, and $\$8 = -1$), and their interaction. We found a significant account remaining $\times$ used (vs. unused) interaction ($b = -8.73$, $SE = 2.63$, $t(1125) = -3.32$, $p < .001$, 95% CI of the difference = $[-13.90, -3.58]$, $\beta = -.14$). An analysis of simple effects revealed that when the gift card is used, participants were more likely to spend \$5 on purchasing the tasty drink as the relative amount remaining in the account *decreased* ($b = -7.99$, $SE = 1.88$, $t(1125) = -4.26$, $p < .001$, 95% CI of the difference = $[-11.68, -4.31]$, $\beta = -.18$). By contrast, when the gift card is unused, the linear trend was not significant ($b = .75$, $SE = 1.84$, $t(1125) = .41$, $p = .69$, 95% CI of the difference = $[-2.87, 4.36]$, $\beta = .017$).

Figure 4. Study 3: The Likelihood of Spending the Money of the Gift Card Across Conditions